Problem 2.75 from 9th edition [Difficulty: 3]

2.75 In a food industry process, carbon tetrachloride at 20°C flows through a tapered nozzle from an inlet diameter D_{in} = 50 mm to an outlet diameter of D_{out}. The area varies linearly with distance along the nozzle, and the exit area is one-fifth of the inlet area; the nozzle length is 250 mm. The flow rate is $Q = 2$ L/min. It is important for the process that the flow exits the nozzle as a turbulent flow. Does it? If so, at what point along the nozzle does the flow become turbulent?

Given: Geometry of and flow rate through tapered nozzle

Find: At which point becomes turbulent

Solution:

Basic equation For pipe flow (Section 2-6) $Re = \frac{\rho \cdot V \cdot D}{\mu} = 2300$ for transition to turbulence

Also flow rate Q is given by $Q = \frac{\pi \cdot D^2}{4} \cdot V$

We can combine these equations and eliminate V to obtain an expression for Re in terms of D and Q

$$Re = \frac{\rho \cdot V \cdot D}{\mu} = \frac{\rho \cdot D}{\mu} \cdot \frac{4 \cdot Q}{\pi \cdot D^2} = \frac{4 \cdot Q \cdot \rho}{\pi \cdot \mu \cdot D} \qquad Re = \frac{4 \cdot Q \cdot \rho}{\pi \cdot \mu \cdot D}$$

For a given flow rate Q, as the diameter is reduced the Reynolds number increases (due to the velocity increasing with A^{-1} or D^{-2}).

Hence for turbulence (Re = 2300), solving for D $D = \frac{4 \cdot Q \cdot \rho}{2300 \cdot \pi \cdot \mu}$

The nozzle is tapered: $D_{in} = 50 \cdot mm$ $D_{out} = \frac{D_{in}}{\sqrt{5}}$ $D_{out} = 22.4 \cdot mm$

Carbon tetrachloride: $\mu_{CT} = 10^{-3} \cdot \frac{N \cdot s}{m^2}$ (Fig A.2) For water $\rho = 1000 \cdot \frac{kg}{m^3}$

$SG = 1.595$ (Table A.2) $\rho_{CT} = SG \cdot \rho$ $\rho_{CT} = 1595 \frac{kg}{m^3}$

For the given flow rate $Q = 2 \cdot \frac{L}{min}$ $\frac{4 \cdot Q \cdot \rho_{CT}}{\pi \cdot \mu_{CT} \cdot D_{in}} = 1354$ LAMINAR $\frac{4 \cdot Q \cdot \rho_{CT}}{\pi \cdot \mu_{CT} \cdot D_{out}} = 3027$ TURBULENT

For the diameter at which we reach turbulence $D = \frac{4 \cdot Q \cdot \rho_{CT}}{2300 \cdot \pi \cdot \mu_{CT}}$ $D = 29.4 \cdot mm$

But $L = 250 \cdot mm$ and linear ratios leads to the distance from D_{in} at which $D = 29.4 \cdot mm$ $\frac{L_{turb}}{L} = \frac{D - D_{in}}{D_{out} - D_{in}}$

$$L_{turb} = L \cdot \frac{D - D_{in}}{D_{out} - D_{in}} \qquad L_{turb} = 186 \cdot mm$$

Problem 2.76 from 9th edition [Difficulty: 2]

2.76 What is the Reynolds number of water at 20°C flowing at 0.25 m/s through a 5-mm-diameter tube? If the pipe is now heated, at what mean water temperature will the flow transition to turbulence? Assume the velocity of the flow remains constant.

Given: Data on water tube

Find: Reynolds number of flow; Temperature at which flow becomes turbulent

Solution:

Basic equation For pipe flow (Section 2-6) $Re = \frac{\rho \cdot V \cdot D}{\mu} = \frac{V \cdot D}{\nu}$

At 20°C, from Fig. A.3 $\nu = 9 \times 10^{-7} \cdot \frac{m^2}{s}$ and so $Re = 0.25 \cdot \frac{m}{s} \times 0.005 \cdot m \times \frac{1}{9 \times 10^{-7}} \cdot \frac{s}{m^2}$ $Re = 1389$

For the heated pipe $Re = \frac{V \cdot D}{\nu} = 2300$ for transition to turbulence

Hence $\nu = \frac{V \cdot D}{2300} = \frac{1}{2300} \times 0.25 \cdot \frac{m}{s} \times 0.005 \cdot m$ $\nu = 5.435 \times 10^{-7} \frac{m^2}{s}$

From Fig. A.3, the temperature of water at this viscosity is approximately $T = 52 \cdot C$

Problem 2.35 [Difficulty: 2]

2.35 SAE 30 oil at 100°C flows through a 12-mm-diameter stainless-steel tube. What is the specific gravity and specific weight of the oil? If the oil discharged from the tube fills a 100-mL graduated cylinder in 9 seconds, is the flow laminar or turbulent?

Given: Type of oil, flow rate, and tube geometry

Find: Whether flow is laminar or turbulent

Solution:

Data on SAE 30 oil SG or density is limited in the Appendix. We can Google it or use the following $\nu = \frac{\mu}{\rho}$ so $\rho = \frac{\mu}{\nu}$

At 100°C, from Figs. A.2 and A.3 $\mu = 9 \times 10^{-3} \cdot \frac{N \cdot s}{m^2}$ $\nu = 1 \times 10^{-5} \cdot \frac{m^2}{s}$

$$\rho = 9 \times 10^{-3} \cdot \frac{N \cdot s}{m^2} \times \frac{1}{1 \times 10^{-5}} \cdot \frac{s}{m^2} \times \frac{kg \cdot m}{s^2 \cdot N} \qquad \rho = 900 \frac{kg}{m^3}$$

Hence $SG = \frac{\rho}{\rho_{water}}$ $\rho_{water} = 1000 \cdot \frac{kg}{m^3}$ $SG = 0.9$

The specific weight is $\gamma = \rho \cdot g$ $\gamma = 900 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times \frac{N \cdot s^2}{kg \cdot m}$ $\gamma = 8.829 \times 10^3 \cdot \frac{N}{m^3}$

For pipe flow $Q = \frac{\pi \cdot D^2}{4} \cdot V$ so $V = \frac{4 \cdot Q}{\pi \cdot D^2}$

$$Q = 100 \cdot mL \times \frac{10^{-6} \cdot m^3}{1 \cdot mL} \times \frac{1}{9} \cdot \frac{1}{s} \qquad Q = 1.111 \times 10^{-5} \frac{m^3}{s}$$

Then $V = \frac{4}{\pi} \times 1.11 \times 10^{-5} \cdot \frac{m^3}{s} \times \left(\frac{1}{12} \cdot \frac{1}{mm} \times \frac{1000 \cdot mm}{1 \cdot m} \right)^2$ $V = 0.0981 \frac{m}{s}$

Hence $Re = \frac{\rho \cdot V \cdot D}{\mu}$

$$Re = 900 \cdot \frac{kg}{m^3} \times 0.0981 \cdot \frac{m}{s} \times 0.012 \cdot m \times \frac{1}{9 \times 10^{-3}} \cdot \frac{m^2}{N \cdot s} \times \frac{N \cdot s^2}{kg \cdot m} \qquad Re = 118$$

Flow is laminar

Problem 8.2

(Difficulty 2)

8.2 Determine the maximum flow rate of air at laminar conditions in a 4-in.-diameter pipe at an absolute pressure of 30 psia and 100°F. Determine the maximum flow rate when (a) the pressure is raised to 60 psia, and (b) the temperature is raised to 200°F. Describe the reason for the differences in flow rates in terms of the physical mechanisms involved.

Find: The maximum flow rate for laminar flow.

Assumption: Air behaves as an ideal gas

Solution: The basic equations are the definition of Reynolds number, the continuity expression, and the ideal gas law

$$Re = \frac{V\rho D}{\mu} = \frac{VD}{\nu}$$

$$m = \dot{\rho}\, A\, V$$

$$\rho = \frac{p}{RT}$$

We have

$$p = 30\ psia = 4320\ \frac{lbf}{ft^2}$$

$$R = 1716\ \frac{lbf \cdot ft}{slug \cdot °R}$$

$$T = 100\ °\text{F} = 560\ °R$$

Thus the density is

$$\rho = \frac{p}{RT} = \frac{4320\ \frac{lbf}{ft^2}}{1716\ \frac{lbf \cdot ft}{slug \cdot °R} \times 560\ °R} = 0.0045\ \frac{slug}{ft^3} = 0.0045\ \frac{lbf \cdot s^2}{ft^4}$$

For the maximum laminar flow we have the Reynolds number at the critical value:

$$Re_{crit} = 2300$$

For this situation

$$D = 4\ in$$

$$\mu = 3.94 \times 10^{-7}\ \frac{lbf \cdot s}{ft^2}$$

The maximum velocity is then:

$$V = \frac{\mu Re}{\rho D} = \frac{3.94 \times 10^{-7}\ \frac{lbf \cdot s}{ft^2} \times 2300}{\left(\frac{4}{12}\ ft\right) \times 0.0045\ \frac{lbf \cdot s^2}{ft^4}} = 0.605\ \frac{ft}{s}$$

The cross section area is:

$$A = \frac{\pi D^2}{4} = \frac{\pi \times \left(\frac{4}{12}\ ft\right)^2}{4} = 0.0873\ ft^2$$

The maximum flow rate:

$$\dot{m} = \rho V A = 0.0045\ \frac{slug}{ft^3} \times 0.605\ \frac{ft}{s} \times 0.0873\ ft^2 = 2.38 \times 10^{-4}\ \frac{slug}{s}$$

If the pressure is raised up to 60 $psi = 8640\ \frac{lbf}{ft^2}$, the density of the air will become:

$$\rho = \frac{8640\ \frac{lbf}{ft^2}}{1716\ \frac{lbf \cdot ft}{slug \cdot °R} \times 560\ °R} = 0.009\ \frac{slug}{ft^3} = 0.009\ \frac{lbf \cdot s^2}{ft^4}$$

The maximum velocity in this case is:

$$V = \frac{\mu Re}{\rho D} = \frac{3.94 \times 10^{-7}\ \frac{lbf \cdot s}{ft^2} \times 2300}{\left(\frac{4}{12}\ ft\right) \times 0.009\ \frac{lbf \cdot s^2}{ft^4}} = 0.302\ \frac{ft}{s}$$

And the maximum flow rate:

$$\dot{m} = \rho V A = 0.009\ \frac{slug}{ft^3} \times 0.302\ \frac{ft}{s} \times 0.0873\ ft^2 = 2.38 \times 10^{-4}\ \frac{slug}{s}$$

The maximum flow rate is the same. The reason is that the density cancels out of the flow rate using Reynolds number:

$$\dot{m} = \rho V A = \frac{\mu Re}{\rho D} \rho A = \frac{\mu Re A}{D}$$

The pressure will not change the viscosity μ (as an assumption)

When the temperature is raised to 200°F, the viscosity decreases. The density also decreases, but we have seen that this has no effect. The viscosity is

$$\mu = 4.49 \times 10^{-7}\ \frac{lbf \cdot s}{ft^2}$$

Thus the flow rate is the ratio of the previous flow rate times the ratio of viscosities

$$\dot{m} = 2.38 \times 10^{-4}\ \frac{slug}{s} \times \frac{4.49 \times 10^{-7}\ \frac{lbf \cdot s}{ft^2}}{3.94 \times 10^{-7}\ \frac{lbf \cdot s}{ft^2}} = 2.71 \times 10^{-4}\ \frac{slug}{s}$$

The reason for the increase is that the viscosity is a function of temperature.

Problem 8.3 from 9th edition [Difficulty: 2]

8.3 For flow in circular tubes, transition to turbulence usually occurs around $Re \approx 2300$. Investigate the circumstances under which the flows of (a) standard air and (b) water at 15°C become turbulent. On log-log graphs, plot: the average velocity, the volume flow rate, and the mass flow rate, at which turbulence first occurs, as functions of tube diameter.

Given: That transition to turbulence occurs at about $Re = 2300$

Find: Plots of average velocity and volume and mass flow rates for turbulence for air and water

Solution:

The basic equations are $Re = \frac{V \cdot D}{\nu}$ $\quad Re_{crit} = 2300$

From Tables A.8 and A.10 $\rho_{air} = 1.23 \cdot \frac{kg}{m^3}$ $\quad \nu_{air} = 1.45 \times 10^{-5} \cdot \frac{m^2}{s}$ $\quad \rho_w = 999 \cdot \frac{kg}{m^3}$ $\quad \nu_w = 1.14 \times 10^{-6} \cdot \frac{m^2}{s}$

For the average velocity $V = \frac{Re_{crit} \cdot \nu}{D}$

Hence for air $V_{air} = \frac{2300 \times 1.45 \times 10^{-5} \cdot \frac{m^2}{s}}{D}$ $\quad V_{air} = \frac{0.0334 \cdot \frac{m^2}{s}}{D}$

For water $V_w = \frac{2300 \times 1.14 \times 10^{-6} \cdot \frac{m^2}{s}}{D}$ $\quad V_w = \frac{0.00262 \cdot \frac{m^2}{s}}{D}$

For the volume flow rates $Q = A \cdot V = \frac{\pi}{4} \cdot D^2 \cdot V = \frac{\pi}{4} \cdot D^2 \cdot \frac{Re_{crit} \cdot \nu}{D} = \frac{\pi \cdot Re_{crit} \cdot \nu}{4} \cdot D$

Hence for air $Q_{air} = \frac{\pi}{4} \times 2300 \times 1.45 \cdot 10^{-5} \cdot \frac{m^2}{s} \cdot D$ $\quad Q_{air} = 0.0262 \cdot \frac{m^2}{s} \times D$

For water $Q_w = \frac{\pi}{4} \times 2300 \times 1.14 \cdot 10^{-6} \cdot \frac{m^2}{s} \cdot D$ $\quad Q_w = 0.00206 \cdot \frac{m^2}{s} \times D$

Finally, the mass flow rates are obtained from volume flow rates

$m_{air} = \rho_{air} \cdot Q_{air}$ $\quad m_{air} = 0.0322 \cdot \frac{kg}{m \cdot s} \times D$

$m_w = \rho_w \cdot Q_w$ $\quad m_w = 2.06 \cdot \frac{kg}{m \cdot s} \times D$

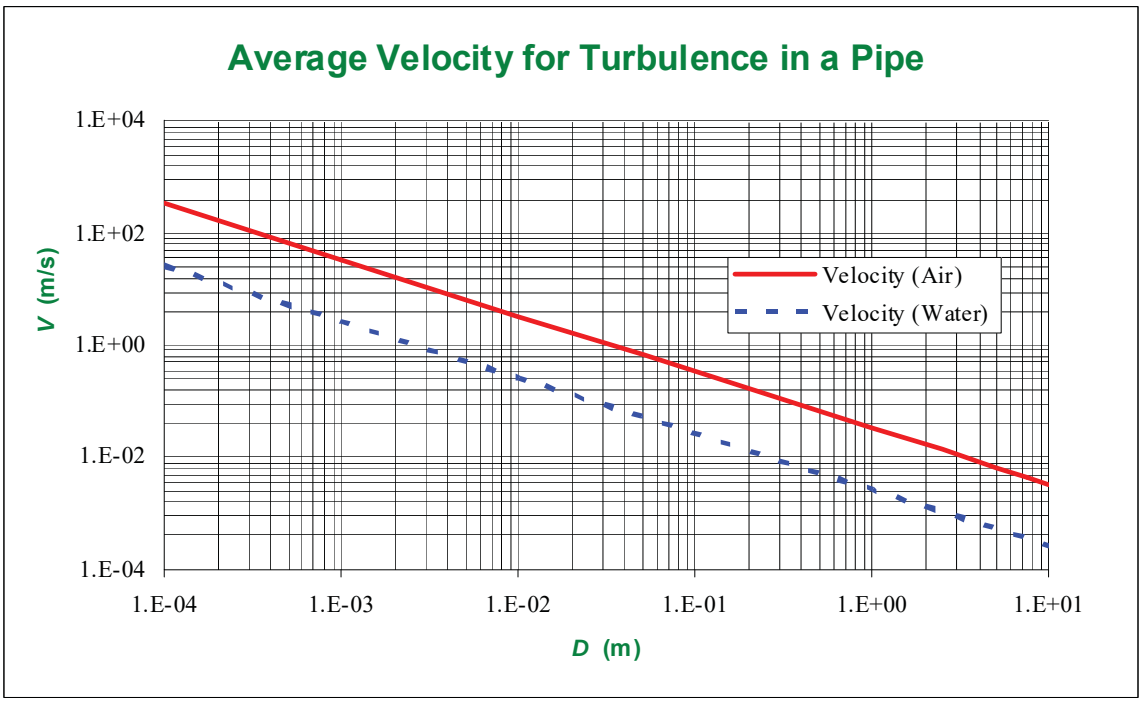
These results can be plotted in *Excel* as shown below in the next two pages

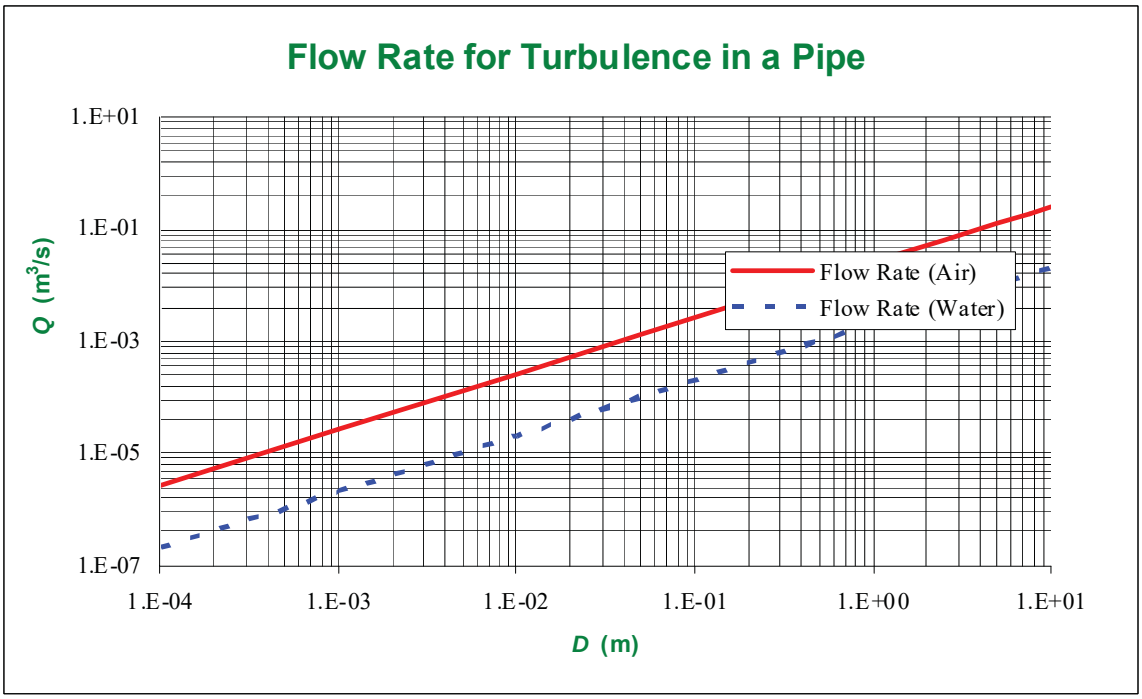
From Tables A.8 and A.10 the data required is

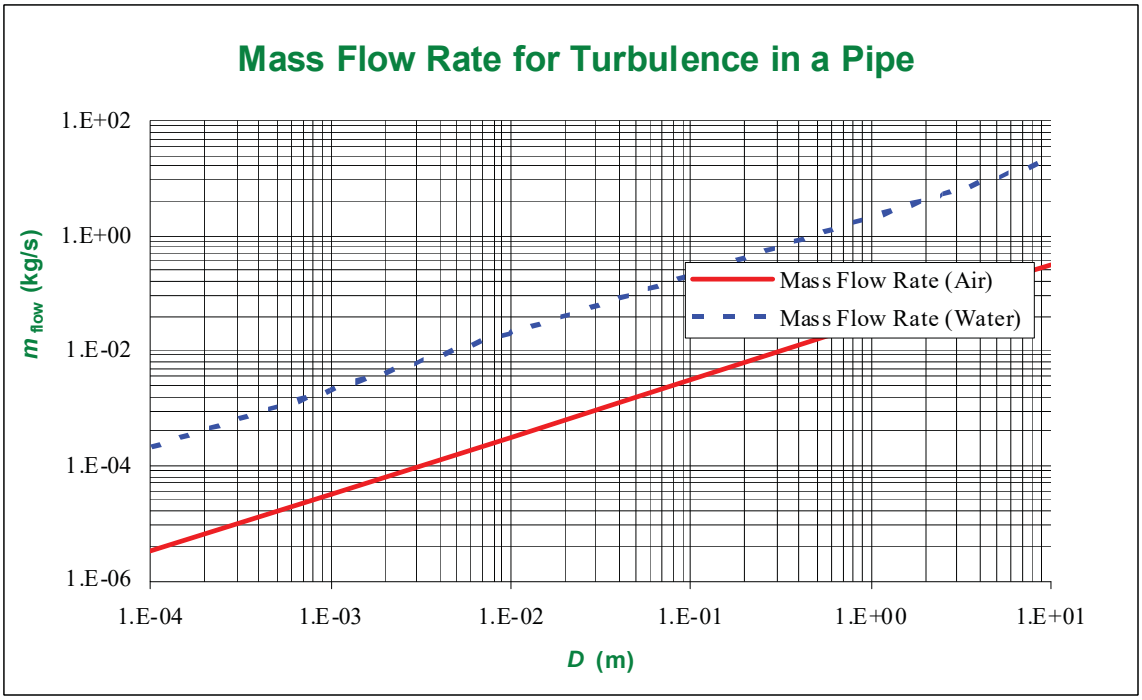
$\Diamond_{air}$ = 1.23 kg/m^3 $\Diamond_{air}$ = 1.45E-05 m^2/s

$\Diamond_{w}$ = 999 kg/m^3 $\Diamond_{w}$ = 1.14E-06 m^2/s

D **(m)**	0.0001	0.001	0.01	0.05	1.0	2.5	5.0	7.5	10.0
V_{air} **(m/s)**	333.500	33.350	3.335	0.667	3.34E-02	1.33E-02	6.67E-03	4.45E-03	3.34E-03
V_{w} **(m/s)**	26.2	2.62	0.262	5.24E-02	2.62E-03	1.05E-03	5.24E-04	3.50E-04	2.62E-04
Q_{air} **(m^3/s)**	2.62E-06	2.62E-05	2.62E-04	1.31E-03	2.62E-02	6.55E-02	1.31E-01	1.96E-01	2.62E-01
Q_{w} **(m^3/s)**	2.06E-07	2.06E-06	2.06E-05	1.03E-04	2.06E-03	5.15E-03	1.03E-02	1.54E-02	2.06E-02
m_{air} **(kg/s)**	3.22E-06	3.22E-05	3.22E-04	1.61E-03	3.22E-02	8.05E-02	1.61E-01	2.42E-01	3.22E-01
m_{w} **(kg/s)**	2.06E-04	2.06E-03	2.06E-02	1.03E-01	2.06E+00	5.14E+00	1.03E+01	1.54E+01	2.06E+01




Flow Rate for Turbulence in a Pipe
1.E+01
1.E-01
1.E-03
1.E-05
1.E-07
Q (m3/s)
Flow Rate (Air)
Flow Rate (Water)
1.E-04
1.E-03
1.E-02
1.E-01
1.E+00
1.E+01
D (m)


Mass Flow Rate for Turbulence in a Pipe
1.E+02
1.E+00
1.E-02
1.E-04
1.E-06
m flow (kg/s)
Mass Flow Rate (Air)
Mass Flow Rate (Water)
1.E-04
1.E-03
1.E-02
1.E-01
1.E+00
1.E+01
D (m)

Problem 8.21

[Difficulty: 3]

8.21 The following tables gives data for the mean velocity in fully developed turbulent pipe flow at Re =50,000 [5]:

$\bar{u}/U$	0.996	0.981	0.963	0.937	0.907	0.866	0.831
y/r	0.898	0.794	0.691	0.588	0.486	0.383	0.280
$\bar{u}/U$	0.792	0.742	0.700	0.650	0.619	0.551	
y/R	0.216	0.154	0.093	0.062	0.041	0.024	

$\bar{u}/U$	0.997	0.988	0.975	0.959	0.934	0.908
y/R	0.898	0.794	0.691	0.588	0.486	0.383
$\bar{u}/U$	0.874	0.847	0.818	0.771	0.736	0.690
y/R	0.280	0.216	0.154	0.093	0.062	0.037

Fit the "power-law" profile for turbulent flow, Eq. 8.22, to the data and obtain a value of n. Plot the data and the correlation on the same graph and verify that Eq. 8.22 is a valid approximation.

Given: Data on mean velocity in fully developed turbulent flow

Find: Trendlines for each set; values of n for each set; plot

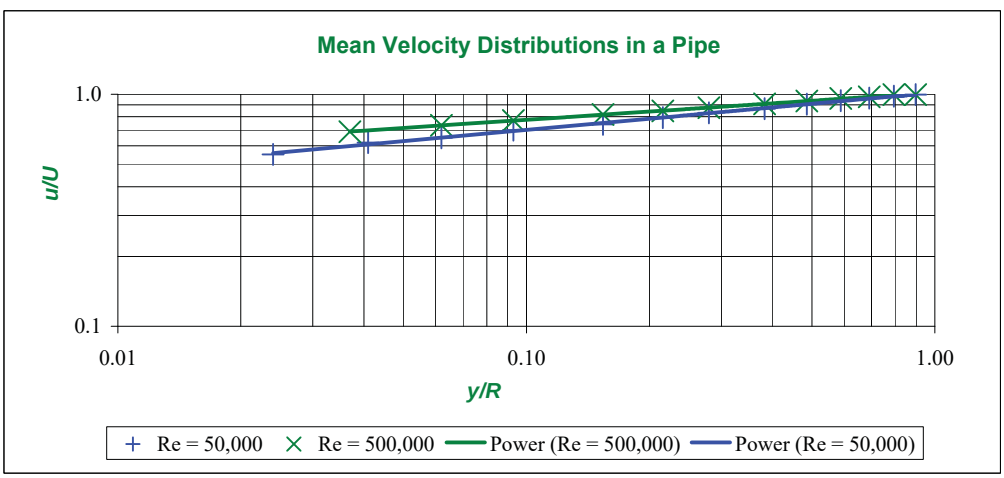
Solution:

y/R	u/U
0.898	0.996
0.794	0.981
0.691	0.963
0.588	0.937
0.486	0.907
0.383	0.866
0.280	0.831
0.216	0.792
0.154	0.742
0.093	0.700
0.062	0.650
0.041	0.619
0.024	0.551

y/R	u/U
0.898	0.997
0.794	0.998
0.691	0.975
0.588	0.959
0.486	0.934
0.383	0.908
0.280	0.874
0.216	0.847
0.154	0.818
0.093	0.771
0.062	0.736
0.037	0.690

Equation 8.22 is

$$\frac{\bar{u}}{U} = \left(\frac{y}{R}\right)^{1/n} = \left(1 - \frac{y}{R}\right)^{1/n}$$



Applying the *Trendline* analysis to each set of data:

At Re = 50,000

$u/U = 1.017(y/R)^{0.161}$

with $R^2 = 0.998$ (high confidence)

Hence $1/n = 0.161$

n = **6.21**

At Re = 500,000

$u/U = 1.017(y/R)^{0.117}$

with $R^2 = 0.999$ (high confidence)

Hence $1/n = 0.117$

n = **8.55**

Both sets of data tend to confirm the validity of Eq. 8.22